

HAT-P-32b

R-0241

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_260106 · created 2026-07-17T05:09:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461046.7671 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.39 +/- 1.99 [%]
Orbital Inclination (inc)	88.98 +/- 1.74
Airmass coefficient 1 (a1)	0.66 +/- 0.34
Airmass coefficient 2 (a2)	0.18 +/- 0.1
Scatter in the residuals of the lightcurve fit is	49.6 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	5.57
Transit Duration (day)	0.1028 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.054 vs 0.1489 (deviation 0.42 σ)
Duration fit vs published	0.1028 d vs 0.1304 d (deviation 2.99 σ)
Mid-time O-C	-7.2 min
Depth SNR	2.2
Residual scatter	49.6 %

Light curve

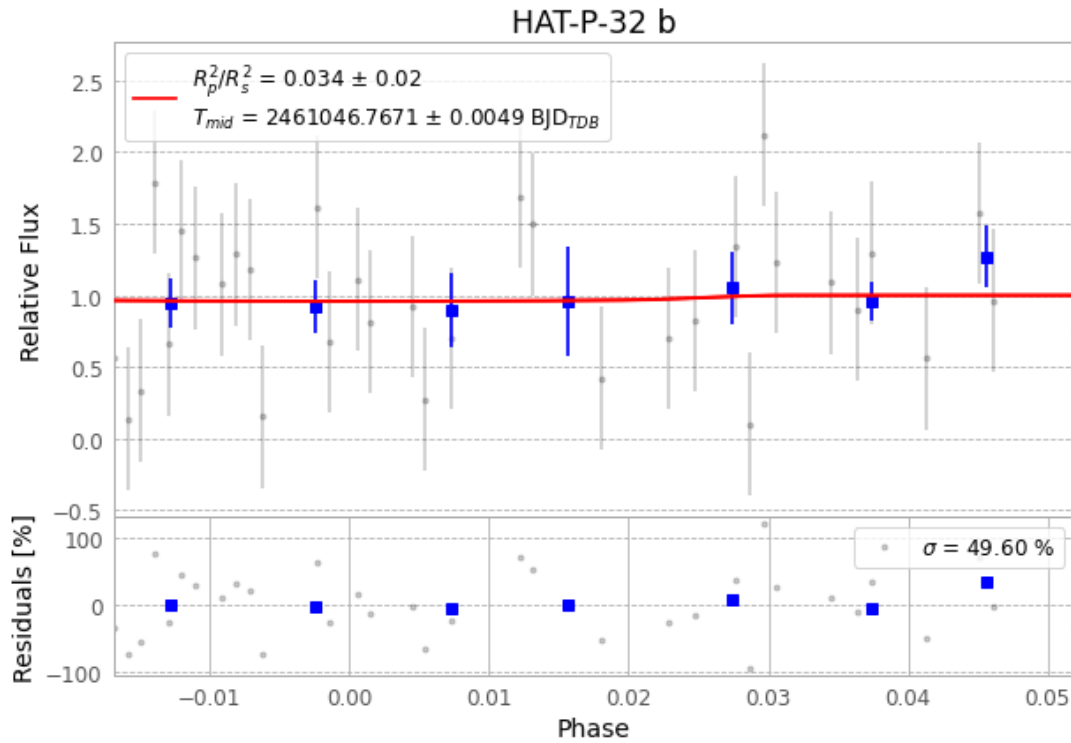


Figure 1 — FinalLightCurve_HAT-P-32 b_2026-01-05.png (SHA-256 decb3961b92c2e96...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 323], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 16b3958fa7ea3c8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), HATP-32260106052715.FITS → HATP-32260106090315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-260106.log (271a4f8d5d39...), FinalLightCurve_HAT-P-32 b_2026-01-05.png (decb3961b92c...), FinalLightCurve_HAT-P-32 b_2026-01-05.csv (cc7a092cfa17...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:09Z.